

ABSTRACT

Children with Autism Spectrum Disorder (ASD) face lifelong social challenges in emotional regulation, social cues and communication. Research has shown that Human–Robot Interaction (HRI) can improve social skills in children with ASD. For children with ASD, robot-assisted–autism-therapy (RAAT) is a novel approach in helping strengthen emotional intelligence and build the bridge for communication. There have been limited implementations of incorporating storytelling with robot interaction. Our objective is to create a social robot to teach children with ASD how to convey and understand emotion. We programmed a robot (simulated TurtleBot) that interacts with users, performing emotionally orchestrated dances based on user feedback. Through storytelling, we've created an interface to teach children emotional behaviors by showing pictures and stories along with emotional responses. The children will be given a quiz, and correct answers will be rewarded with expressive movements from the TurtleBot, providing immediate feedback. Our current implementation creates an interactive and expressive robot that is capable of responding to autistic children through voiced storytelling and motion-based feedback. We've created a seamless interaction between our ROS environment and our story-telling GUI that can be transferable to a physical environment. By collaborating with social workers, parents, and caregivers, we will collect insights on the robot's impact and areas for improvement. Building on these findings, we plan to integrate a more advanced robot capable of supporting complex interactions and incorporating emotional recognition capabilities.

MOTIVATION

Children with Autism Spectrum Disorder live with social challenges in emotional regulation, social cues and communication. RAAT therapy has been proven to strengthen social skills [2]

- Creating an interactive and motivational learning tool for children with ASD to address communication challenges and specifically to help them understand and express emotions
- Through the use of a social robot, this project seeks to provide an innovative approach to improving emotional regulation and social skills, offering a motivating environment for children to learn and interact.

BACKGROUND

- Research on the use of robots for storytelling has shown that combining emotional gestures, sound effects, and background music can enhance the storytelling experience, induce emotions, and improve children's emotional responses [1][3]
- Research shows that using robot-assisted therapy can motivate children with ASD to participate in activities [2]
- By combining interactive storytelling with emotional cues and utilizing engagement measurement tools, these interventions can provide tailored support to meet the unique needs of each child [3][5]

APPROACH

- We developed a child-friendly graphical user interface (GUI) to guide children through interactive storytelling activities using simulated social robot.
- The simulated robot in the Gazebo environment provides responsive feedback based on the input.
- Our system is tailored for children with ASD to engage in narratives and emotional cues to encourage learning.

PROTOTYPE



Figure 1: Flashcard screen

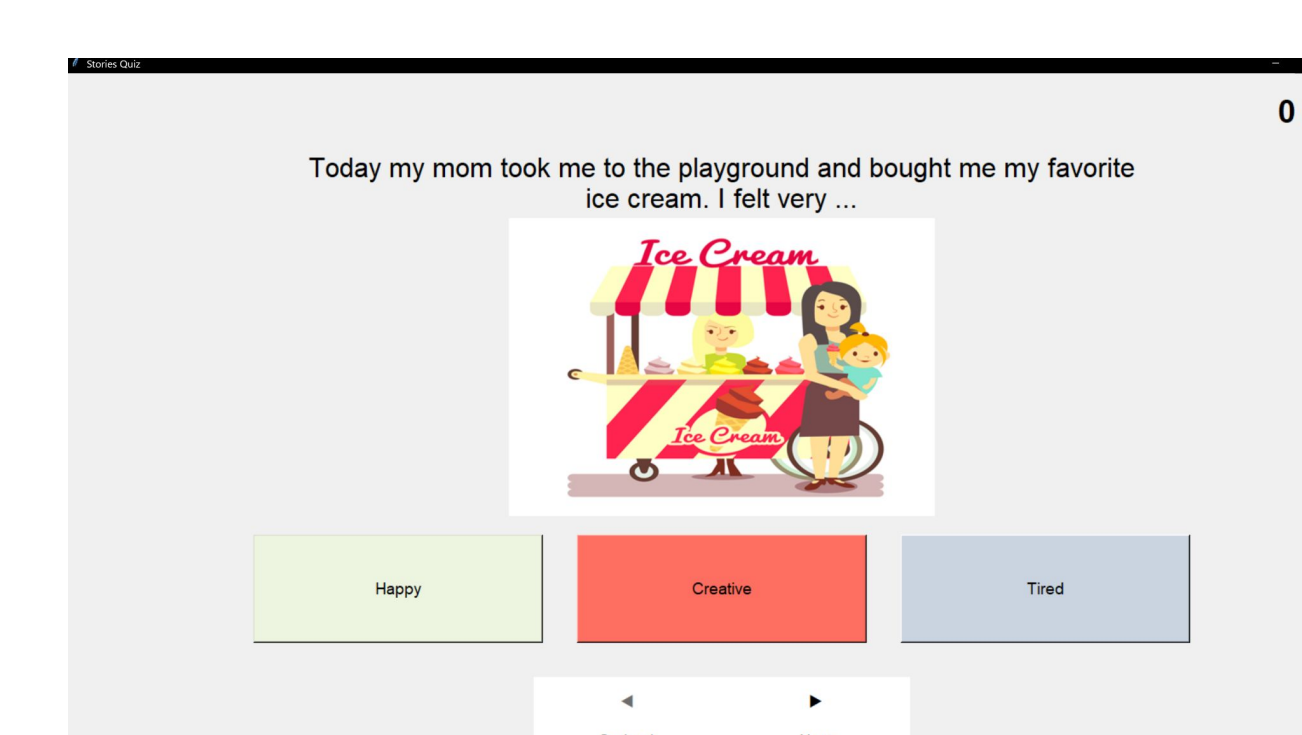


Figure 2: Story Quiz Screen



Figure 3: TurtleBot Robot

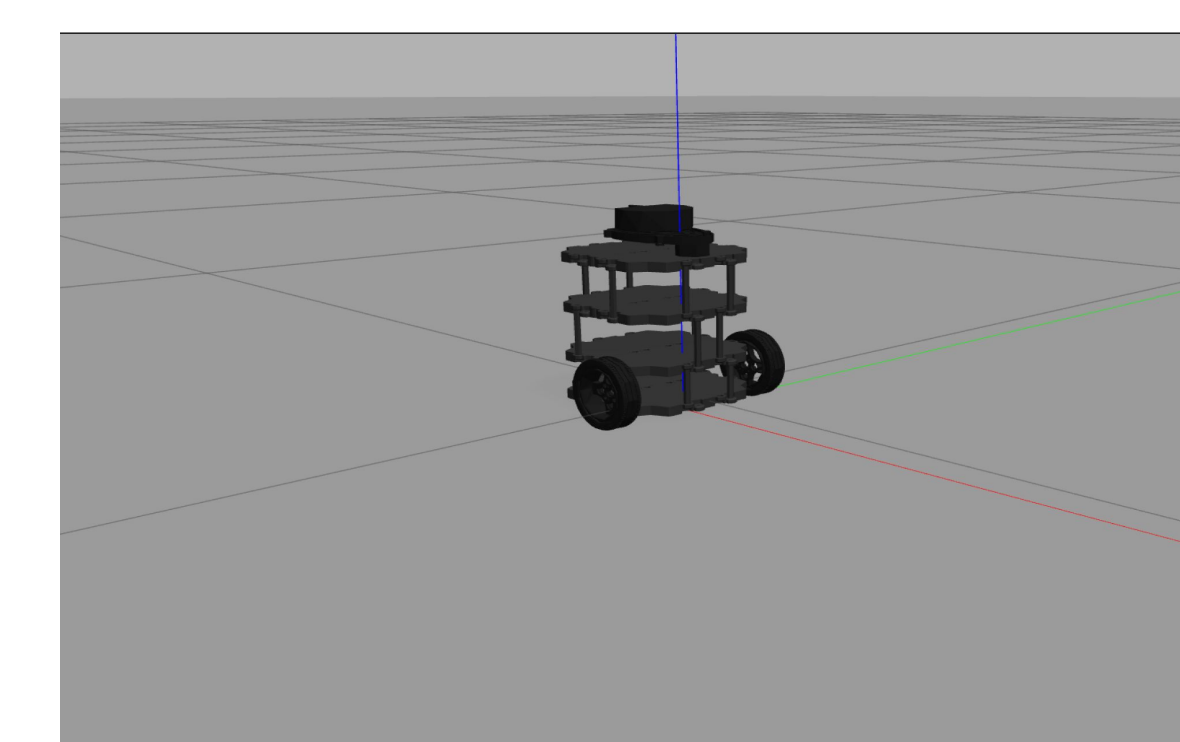


Figure 4: Standalone ROS Environment

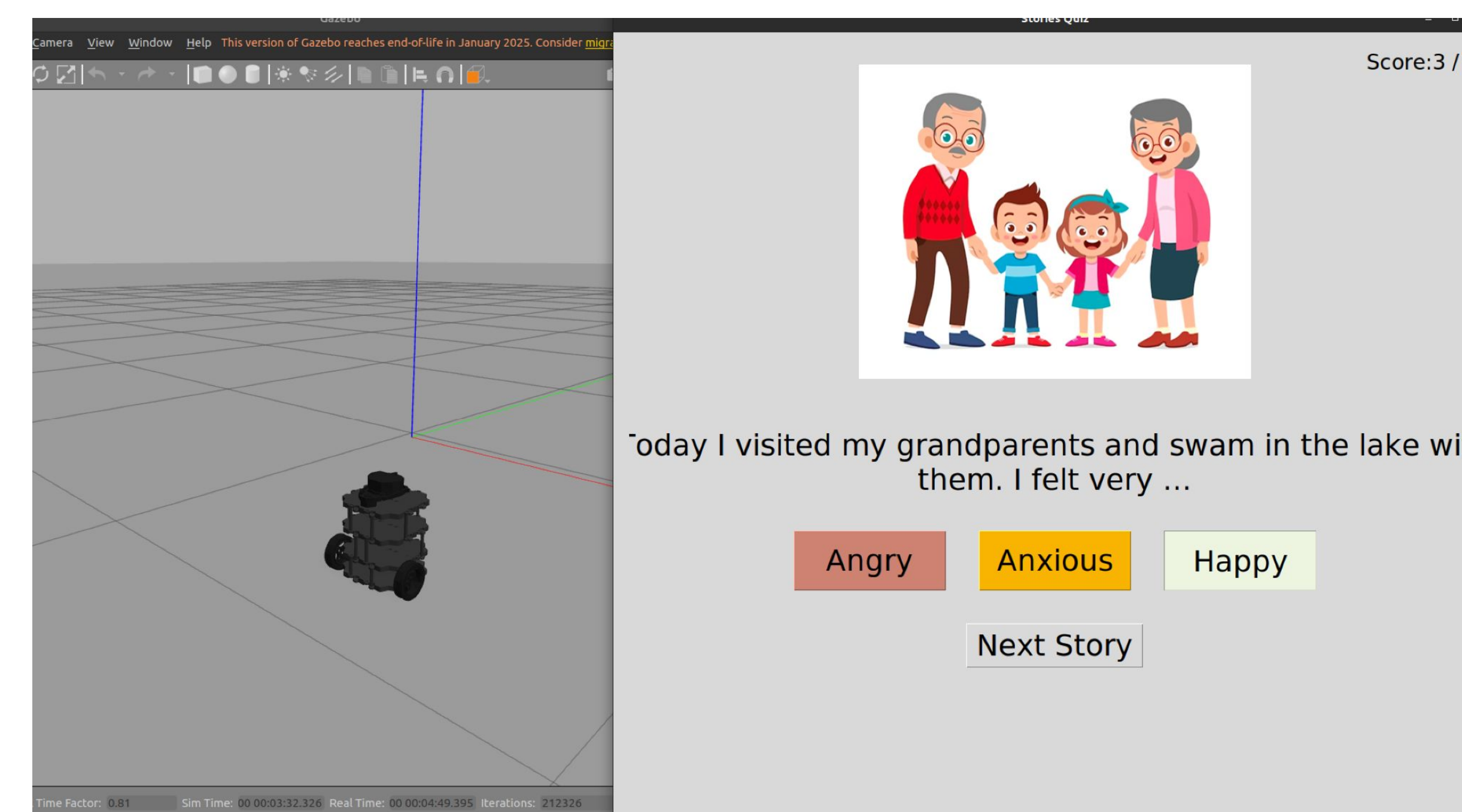


Figure 4: Virtual environment moving along with Story Quiz

CONCLUSION AND FUTURE DIRECTION

Our goal with this project is to build the foundation for an interactive storytelling platform where children can learn to recognize and express emotions through engaging narratives, robot feedback, and visual cues. By combining story-based learning with emotion choices and real-time feedback, we aim to support children's emotional development in a fun and supportive environment.

- **Emotional Recognition:** Incorporate speech and facial emotion recognition to allow the system to adapt dynamically to a child's emotional state, enabling more personalized support.
- **Customizable Stories:** Add more diverse and relatable stories that are personalized for each child
- **Advanced Robot Interaction:** transition from a simulated TurtleBot to a more advanced robot that supports complex gestures, facial displays, and multi-modal communication to deepen interaction quality.
- **Enhanced Accessibility:** Add more languages and other accessibility features to support a diverse range of learning needs.

REFERENCES

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ACKNOWLEDGEMENTS

- Faculty Research Advisor: Dr. Mohammad Q. Azhar
 - AI Tech Innovation Hub,
 - Tech Learning Community
 - Professor Raj Korpan, Jason Shan
- Questions/Comments: Dr. Azhar mazhar@bmcc.cuny.edu